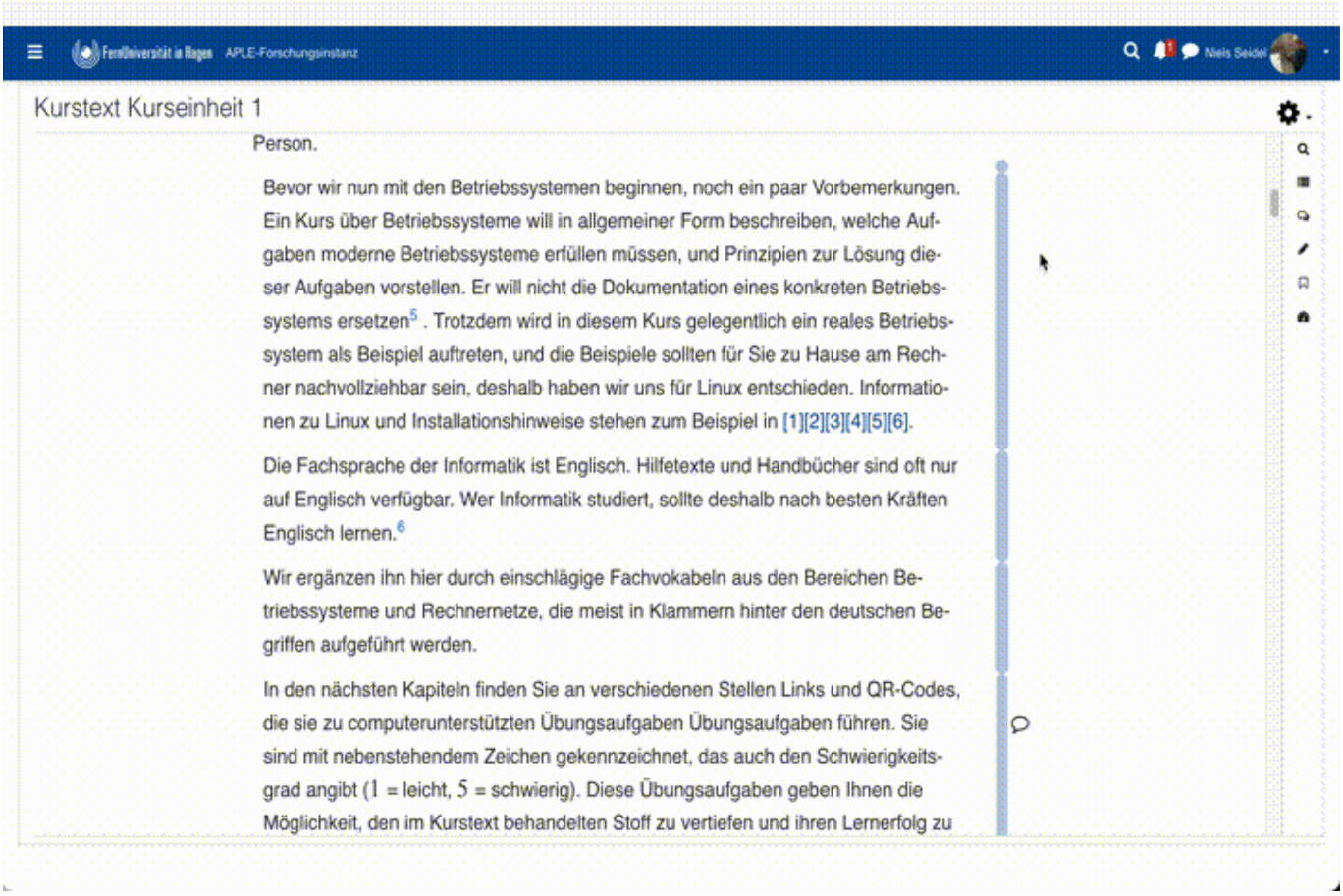

Longpage

Longpage is a Moodle activity plugin for reading long texts in online courses paired up with functions to navigate, annotate, test reading comprehension, and co-read with other students.

Longpage (mod_longpage) is based on the Page or mod_page module which is part of the standard installation of Moodle. *Page* by itself simply allows for providing texts included in a Moodle page with no further functionality attached. *Longpage* can be seen as an extension of *Page* with is to be used as a replacement of *Page* for providing users with extra functionality.

Maintained? yes | github NOT_FOUND | github 404 | issues repo not found | License GPLv3

Tested Moodle versions 3.10, 3.11 | PHP ~8.0.29 | NPM ~10.1.0 | node.js ~18.17.0 | vue.js 2

stability mature | years in productive use 4 | used in unique courses 5

collects clickstream data yes | collects scroll data yes | collects mouse data no | collects audio data no | collects video data no | data shared with vendor no

AI methods recommender system | Base model none | Feature is optional yes | Human in the loop no

Key Features • How To Use • Download • Credits • Related • Citation • License

Key Features

The main features *Longpage* adds on top of *Page* are:

Foster reading of longer text:

- reading-friendly design
- renders any HTML including images, videos, tables, and mathematical formulas (expressed in TeX notation).
- automatic creation of interactive table of contents showing current reading position and allowing users to jump among sections of the text provided
- full-text search
- saves the last scroll position
- approximates reading time per section and the overall document
- visualises reading progress indicating how often a paragraph has been read

Support elaboration of text:

- highlight text in four different colors and underline text in red or black
- set a bookmark for a selection of text to remember that selection and jump back to it
- add personal notes

Enabling co-reading:

- share notes among users publicly and anonymously,
- thread-based discussions from notes shared and contribute to them, e.g., answer a question by another user,
- subscribe to notifications regarding changes in threads, especially new posts/notes by other users,
- bookmark notes,
- like notes,
- determination if post has been read by user and ability to mark notes as read or unread,
- filter threads and posts by a set of filters regarding content, author, number of likes, status (read, liked, subscribed to, bookmarked), time created or last modified,
- sort threads by time last modified or created, position of anchor in text, relevance based on novelty, collaborative filtering and reading progress of user.

Learning Analytics included

- collection of clickstream data as well as text highlighting, bookmarks, notes, discussions, and search terms
- capturing reading activities using the Intersection Oversever to detect text elements visible on screen
- visualisation of reading progress as a vertical bar right next to a paragraph

Roadmap and Limitations

Roadmap

- export to document to PDF including comments and highlighted text
- andvanced reading comprehension support

Limitations

- Longpage supports HTML instead of PDF

How To Use

To clone and run this application, you'll need Git and Node.js (which comes with npm) installed on your computer. From your command line:

```
# Clone this repository
1. Clone  the repository to /your-moodle/mod/
$ git clone XXX

# Rename the folder to 'longpage'
$ mv mod_longpage longpage

# Go into the repository
$ cd longpage

# Install dependencies
$ cd vue
$ npm install

# Build the plugin by transpiling the vue code into javascript
$ npm run build

# Open the page https://<moodle>/admin/index.php?cache=1 and follow the
install instructions for the plugin or
$ php admin/cli/uninstall_plugins.php --plugins=mod_longpage --run

# To install the *Longpage* plugin afterwards, copy the repository
downloaded in the 1. step into the `mod` folder in the folder your Moodle
installation is located in replacing the current `mod/longpage` folder
containing the regular *Page* plugin. Now, login to your Moodle running as
an administrator. The install/update GUI should open automatically. Just
follow the steps the GUI presents to you and you should have installed the
*Longpage* plugin successfully afterwards. As an alternative to using the
GUI for installation, you can also run the update script from within the
folder of your Moodle installation:
$ php admin/cli/upgrade.php

# Open a Moodle course of you choice and add longpage as an activity to
your course.
```

Before you can install the plugin you should have a proper Moodle installation running. See here for a tutorial on how to install Moodle. Pay special attention to enabling Cron since it is necessary for some functionality of the Longpage plugin (notifications on activity of other users, calculation of relevance of annotations/threads/posts for a user). The plugin has been developed with Moodle version 3.9 since it represent the current LTS but it has also been tested with other version between 3.5 and 3.10 so it should work with them as well. We cannot make any statements about other versions since we have not tested it. Just give it a try 😉

Troubleshooting

- disable caching in Moodle's config.php (`cachejs=false;`)
- remove, install or reinstall node_modules (npm install)
- moodle: purge caches and disable caching (use search: 'cache' on moodle site administration)
- If there is a javascript error that app-lazy.js could not be loaded, try to create a symbolic link from app-lazy.min.js to app-lazy.js. For Windows users:

```
mklink mod\longpage\amd\src\app-lazy.min.js mod\longpage\amd\build\app-lazy.min.js
```

- In case of an error like "ERR_OSSL_EVP_UNSUPPORTED" run `export NODE_OPTIONS=--openssl-legacy-provider`
- Compile errors like broken strings may be caused by wrong version of node.js. Install node v18 (check version `node --version`) and rerun `$ npm run build`.

Plugin Settings TODO incl. screenshot

Embedding Reading Comprehension Questions In order to use embedded reading comprehension questions, you will need to install the "Embed questions filter", "Embed question atto button" and "Embedded questions progress" Moodle plugins. Documentation on how to embed questions with the plugin can be found here.

Basically, with these plugins, you have to add a cryptic code for each question at the location in the text in the longpage content where the question should appear in the sidebar when scrolling over the text.

You will also need to enable the reading comprehension feature in the settings.

Download

You can download the latest installable version of Longpage for Moodle 3.11 and Moodle 4.x.

Getting into Development

To contribute to the plugin you should study the Moodle documentation on plugin development deeply. The plugin is very similar to a regular Activity plugin. The main difference regards the client or asynchronous javascript modules (AMD) (`amd` directory). Instead of creating javascript files inside `amd` in require.js format from scratch, working with Grunt etc., the main part of the client which is a SPA for reading, annotating and navigating the text, is developed with Vue.js v3. The files are located inside the `vue` directory which is

where you mainly develop the client like you would develop a regular SPA with Vue.js. Sadly, you do not have hot reload like you are maybe used to from other projects with Vue.js. Instead you run

```
npm run watch
```

inside the `vue` directory. From then on Webpack, the bundler used for this project, is going to watch all the files in `vue` and its direct and indirect subdirectories. Whenever a file changes, a new (developmental) version of `amd/app-lazy.min.js` which is the bundle that contains all html, css and javascript of the SPA. You then have to reload the page inside the browser if you are currently on it to view changes you made to the SPA.

When deploying to production, you run

```
npm run build
```

to let Webpack bundle up a production ready version of the plugin. IMPORTANT: Do not use `npm run watch` to build for production since it lacks some optimizations that Webpack applies, i.e. in order to make the bundle smaller in size.

Emailware

Longpage is an emailware. Meaning, if you liked using this plugin or it has helped you in any way, I'd like you send me an email at niels.seidel@fernuni-hagen.de about anything you'd want to say about this software. I'd really appreciate it!

Credits

This software uses the following open source packages:

- vue.js
- vuex
- vue-router
- node.js

Related

- mod_page - Moodle core plugin
- mod_pdfannotator - xxx

Citation

Cite this software:

TODO: ZENODO

Research articles and datasets about Longpage:

- Menze, D., Seidel, N., & Kasakowskij, R. (2022). Interaction of reading and assessment behavior. In P. A. Henning, M. Striewe, & M. Wölfel (Eds.), DELFI 2022 – Die 21. Fachtagung Bildungstechnologien der Gesellschaft für Informatik e.V. (pp. 27–38). Gesellschaft für Informatik. https://doi.org/10.18420/delfi2022-011
- Seidel, N., & Menze, D. (2022). Interactions of reading and assessment activities. In S. Sosnovsky, P. Brusilovsky, & A. Lan (Eds.), 4th Workshop on Intelligent Textbooks, 2022 (pp. 64–76). CEUR-WS. http://ceur-ws.org/Vol-3192/
- Menze, D., & Seidel, N. (2022). Support for Reading Comprehension in Digital Course Texts. In M. Mandausch & P. A. Henning (Eds.), Proceedings of DELFI Workshops 2022 (pp. 105–116). https://doi.org/10.18420/delfi2022-ws-21
- Seidel, N., & Menze, D. (2022). Data and Analysis of Reading and Assessment Activities in Moodle. Zenodo. https://doi.org/10.5281/zenodo.7300070

You may also like...

- format_serial3 - Learning Analytics Dashboard for Moodle Courses
- mod_usenet - Usenet client for Moodle
- local_ari - Adaptation Rule Interface
- hypercast
- tex2html - Convert LaTeX to HTML longpage format

License

GNU GPL v3 or later

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